English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2021/2022 Test 1 — *Your grade is 2 + the number of points. Time allowed: 150 minutes. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$. *Unless otherwise stated, a random choice is uniform.*

Problem 1. (1 point) One box contains an ordinary six-sided die, another a fair eight-sided die labelled 1 through 8. Ivan chooses a box at random, rolls its die and announces the result. As a function of the announced number, what is the probability that it came from the ordinary die?

Problem 2. (1 point) Approximate the numbers of first-, second- and third-year students by 600, 400 and 200, respectively. Their probabilities of attending a class are 80%, 60% and 25%. Choose two random students who attended classes. Given that they are from different years, what is the probability the more senior is in year 3?

Problem 3. An examination has 20 questions, each with four possible answers. A student knows ten answers and answers them correctly, and guesses the other ten at random. Let X count correct answers. Find:

1. (0.3 points) the distribution of X;
2. (0.3 points) $\mathbb{P}(X > 15)$;
3. (0.4 points) $\mathbb{E}X$ and DX. Full credit requires explicit answers; calculations can avoid binomial coefficients.

Problem 4. A message consists of eight bits. Noise between servers causes each bit to be transmitted incorrectly with probability p.

0	1	0	0	0	1	1	1

1. (0.5 points) What is the probability that a message is transmitted correctly between directly connected servers A, B? What is the expected number of correct bits in the final message?
2. (0.5 points) Answer the same questions if the message passes exactly once through each of $n = 3$ intermediate servers. Bonus: what happens as $n \to \infty$?